

# Action Theory

## Constraint-Freedom-Roadblock as an Action-Fundamental Theory of Reality

**Research status:** speculative working theory; not established physics.

**Core hypothesis:** The primitive structure of reality is not a pre-existing universe containing matter, fields, space and time. The primitive structure is an algebra/network of admissible actions and their composition.

$$\boxed{\text{fundamental reality} = (\mathfrak{A}, \circ)}$$

where  $\mathfrak{A}$  is the set of elementary actions and  $\circ$  is their physical composition rule.

### Core CFR definitions

For a partial action history  $h$ , let

$$\mathcal{N}(h) = \{\alpha : \alpha \circ h \text{ is physically admissible}\}.$$

Then:

$$\boxed{\text{Constraint} = \text{restriction defining } \mathcal{N}(h)}$$

$$\boxed{\text{Freedom} = F(h) = \dim \mathcal{N}(h)}$$

and

$$\boxed{\text{Roadblock } R(h) \iff \mathcal{N}(h) = \emptyset.}$$

A Roadblock terminates a particular action channel, not reality itself. A chemical reaction can finish while other processes remain possible.

### Emergent ontology

A state is an equivalence class of histories with equivalent future possibilities:

$$h_1 \sim h_2 \iff \mathcal{N}(h_1) \cong \mathcal{N}(h_2).$$

Therefore

$$\boxed{\text{state} = [h]}$$

and a persistent object is a stable equivalence class of action histories.

Proposed hierarchy:

$$\text{action} \rightarrow \text{history} \rightarrow \text{state} \rightarrow \text{stable object/matter}$$

## Quantum theory

Assign each action a complex amplitude:

$$Z(\beta \circ \alpha) = Z(\beta)Z(\alpha).$$

For a history,

$$Z(\gamma) = e^{iS[\gamma]/\hbar}.$$

The transition amplitude is

$$K(X_f, X_i) = \sum_{\gamma: X_i \rightarrow X_f} e^{iS[\gamma]/\hbar}.$$

In the semiclassical limit,

$$\delta S = 0,$$

recovering stationary-action mechanics.

With a complex linear state space and continuous composition,

$$i\hbar \frac{d}{dt} |\psi\rangle = H |\psi\rangle.$$

For a nonrelativistic particle,

$$i\hbar \frac{\partial \psi}{\partial t} = \left[ -\frac{\hbar^2}{2m} \nabla^2 + V \right] \psi.$$

**Important unresolved point:** CFR has not yet derived why the fundamental action algebra must use complex amplitudes, the Hilbert-space norm, unitarity, and the Born rule.

## Entanglement

Independent action structures factor:

$$\mathcal{A}_{AB} = \mathcal{A}_A \otimes \mathcal{A}_B.$$

Entangled structures do not:

$$\boxed{\mathcal{A}_{AB} /} = \mathcal{A}_A \otimes \mathcal{A}_B.$$

CFR interpretation:

$$\boxed{\text{entanglement} = \text{nonfactorizability of fundamental action possibilities.}}$$

This allows spatially separated emergent systems to remain fundamentally related without requiring controllable faster-than-light signalling.

## Space and time

Time is hypothesized to emerge from causal ordering:

$$\alpha \prec \beta$$

when  $\beta$  is an admissible continuation of  $\alpha$ .

Space is hypothesized to emerge from the geometry of mutually independent action directions.

If the large-scale action network develops a Lorentzian metric,

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2,$$

then the null boundary gives

$$ds^2 = 0$$

and therefore

$$|\mathbf{v}| = c.$$

Thus

$$\boxed{c = \text{propagation rate of the null boundary of emergent action accessibility.}}$$

**Unresolved:** CFR has not yet derived the Lorentzian signature or the fact that the macroscopic dimension is four.

## Fields

A field is proposed to be a collective coarse-grained pattern of action relationships.

For a U(1) sector, introduce link variables

$$U_{ij} \in U(1)$$

with

$$U_{ij} \rightarrow e^{i\alpha_i} U_{ij} e^{-i\alpha_j}.$$

In a continuum limit,

$$U_{ij} \rightarrow \exp \left( ie \int_i^j A_\mu dx^\mu \right),$$

giving

$$A_\mu \rightarrow A_\mu + \frac{1}{e} \partial_\mu \alpha.$$

The curvature becomes

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

and the effective action becomes Maxwell theory.

CFR interpretation:

$$\text{electromagnetic field} = \text{curvature of an internal action connection.}$$

The Higgs field is proposed as a collective order parameter of the vacuum action network:

$$V(\Phi) = -\mu^2 |\Phi|^2 + \lambda |\Phi|^4.$$

The observed nonzero vacuum configuration would correspond to a stable phase of the underlying action network.

## Matter

Matter is hypothesized to be a stable localized action pattern satisfying

$$\frac{\delta S_{\text{eff}}}{\delta \Phi} = 0$$

with a stability condition

$$\delta^2 S_{\text{eff}} > 0$$

in physical, non-gauge directions.

Thus

$$\text{matter} = \text{stable localized action pattern.}$$

A particle's mass would be its invariant rest energy,

$$mc^2 = E,$$

and its quantum numbers would arise from invariants of the underlying action structure.

A major CFR goal is therefore:

derive the particle spectrum from stable action solutions.

## Gravity

In a coarse-grained regime, suppose the action network produces an emergent metric  $g_{\mu\nu}$ .

Then the effective gravitational action is expected to reduce to

$$S_{EH} = \frac{c^3}{16\pi G} \int d^4x \sqrt{-g} (R - 2\Lambda),$$

giving

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.$$

CFR interpretation:

gravity = dynamical geometry of physically accessible action.

**Unresolved:** CFR has not derived the Einstein-Hilbert action uniquely from the microscopic action algebra.

## Freedom Geometry

The main genuinely new CFR object is a metric on the space of physically admissible continuations:

$$\mathbf{G}_{AB}.$$

Its local freedom volume is

$$\mathcal{V}_F = \sqrt{|\det \mathbf{G}|}.$$

A regular action regime satisfies

$$\mathcal{V}_F > 0.$$

A Roadblock candidate is

$$\det \mathbf{G} = 0$$

or a change in rank.

The central conjecture is therefore:

ordinary physics = evolution on regular strata of action space

while

new physics = transition at a Roadblock where regularity fails.

## Candidate new quantum physics

A phenomenological test model is

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + \Gamma_F \mathcal{D}_Q[\rho],$$

with

$$\Gamma_F = \gamma_* \left[ \frac{\mathcal{R}_F - \mathcal{R}_c}{\mathcal{R}_c} \right]_+^p.$$

The intended signature is a threshold:

$$\mathcal{R}_F < \mathcal{R}_c \Rightarrow \Gamma_F = 0,$$

followed by a new regime when

$$\mathcal{R}_F \geq \mathcal{R}_c.$$

This should not be replaced by a simple mass threshold; large-mass matter-wave interference has already been demonstrated experimentally.

## Candidate new gravitational physics

Introduce a CFR correction

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} + \lambda_F \mathcal{K}_{\mu\nu}[\mathfrak{A}].$$

Ordinary regimes should satisfy

$$\mathcal{K}_{\mu\nu} \approx 0.$$

A Roadblock or strongly distorted freedom geometry could produce

$$\mathcal{K}_{\mu\nu} \neq 0. \quad =$$

The key research target is to derive  $\mathcal{K}_{\mu\nu}$  from the microscopic action algebra rather than assume it.

A possible application would be a controlled modification of gravitational response. This is the mathematical route by which a CFR “antigravity engine” could exist, but no such technology is currently established.

## Required derivations

The theory is incomplete until it can derive:

1. Complex quantum amplitudes and the Born rule.
2. Lorentzian signature  $(-+++)$ .
3. Four macroscopic dimensions.
4. The gauge group  $SU(3) \times SU(2) \times U(1)$ .
5. The observed particle spectrum.
6. The Freedom Geometry  $\mathbf{G}_F$ .
7. A nonzero Roadblock correction.
8. A quantitative prediction distinguishable from standard QM and GR.

The crucial criterion is:

$$\text{one microscopic action algebra} \Rightarrow QM + GR + SM + \text{new prediction.}$$

## Research principle

Do **not** count a reinterpretation as a derivation.

Do **not** insert GR, QM, the Standard Model, collapse, antigravity, dark matter or a gauge group and then call them consequences.

Every ingredient must be classified as:

derived,   assumed,   effective,   or predicted.

## First computational target

Construct the smallest finite action category/graph with:

$$(\mathfrak{A}, \circ, C, Z, \mathcal{G})$$

where

- $\mathfrak{A}$  = elementary actions,
- $\circ$  = composition,
- $C$  = admissibility constraints,
- $Z$  = action amplitudes,
- $\mathcal{G}$  = representation/gauge redundancy.

Then determine, without inserting spacetime or particles by hand, whether the model produces:

|                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------|
| quantum interference,   entanglement,   gauge invariance,   a causal structure,   stable excitations,   Roadblock |
|-------------------------------------------------------------------------------------------------------------------|

The decisive question is:

|                                                                                                            |
|------------------------------------------------------------------------------------------------------------|
| <b>Can one action algebra force the observed low-energy laws and simultaneously predict something new?</b> |
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That is the central research problem of Action Theory.